A SUBSPACE MODEL OF SPEECH-WATERMARK SURVIVAL UNDER ANALYSIS–RESYNTHESIS

Weiping Yan

University of Minnesota, Twin Cities, Minneapolis, MN, USA

ABSTRACT

Speech watermarks are widely reported to fail when audio is regenerated by a vocoder or neural codec. We give this phenomenon a restricted but precise model: a blind launderer is an *analysis–resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$. At full generality we prove only what is true: a data-processing inequality (laundering cannot increase the Chernoff detection exponent), and an *exact-erasure condition* — if embedding leaves the analysis output unchanged a.s., the laundered laws coincide and no detector beats chance. In a linear-Gaussian surrogate we obtain closed forms: the channel transmits exactly the equivalence class of the perturbation modulo $\ker \mathcal{A}$, with surviving exponent $\frac{1}{8}\|P\delta\|^2$; a Gaussian *achievable lower bound* R_{LB} (not a capacity) covers blind embedders. The model's empirical content is a measurable, channel-relative *survival predictor*: each attacker's own analysis sensitivity. On LibriSpeech ($[pending]$ speaker-disjoint test clips, thresholds from $[pending]$ independent calibration clips), across spectral inversion, two neural vocoder pipelines, three neural codecs, and self voice-conversion, the predictor tracks held-out post-attack separability (Spearman $[pending]$, perm. $p = [pending]$), outperforming SNR and spectral-centroid controls. Deployed watermarks (AudioSeal, WavMark, SilentCipher) at matched perceptual quality fail *operationally* under resynthesis — score sign/order inversions under their original operating rules — while attack-conditioned separability often remains; we report both, and never conflate them.

Index Terms— audio watermarking, generative AI, provenance, data-processing inequality, hypothesis testing

1. INTRODUCTION

Provenance watermarking embeds a key-dependent signal in speech so a detector can later flag synthetic or traced audio [1, 2, 3]. A consistent empirical finding is fragility to *regeneration*: passing watermarked audio through a vocoder, neural codec, or voice conversion drives deployed detectors toward failure [4, 5]. The community's response is to embed in representations that resynthesis preserves — codec latents (Latent-Mark [6]), speech-feature-aligned marks [7], speaker-latents robust to zero-shot VC [8], and semantic image watermarks before them [9, 10, 11]. The *invariant-alignment idea is thus established practice, not our contribution.*

What the literature lacks is a stated model in which the collapse and its exceptions can be derived and *predicted*. Empirical papers report which schemes die under which attacks; the pattern — AudioSeal survives EnCodec yet fails under mel-inversion; marks with more energy in preserved components survive longer — is left as observation. This paper supplies the missing object: a channel model with exactly-scoped guarantees, and a *measurable* quantity that predicts survival out-of-sample.

Contributions. (i) A general proposition for blind analysis–resynthesis channels: a Chernoff data-processing inequality, and a sufficient condition for *exact* erasure (§2). Nothing stronger is claimed at this generality. (ii) A linear-Gaussian surrogate where everything is closed-form: the channel transmits the perturbation's class in $\mathbb{R}^n / \ker \mathcal{A}$; mixed perturbations survive *partially* with exponent $\frac{1}{8}\|P\delta\|^2$ (§3). (iii) A Gaussian achievable lower bound R_{LB} for blind embedders — explicitly *not* a watermarking capacity: informed embedding [12, 13] can exceed it and no rate converse is claimed (§3). (iv) A channel-relative survival predictor — each attacker's own analysis sensitivity — validated on *held-out, speaker-disjoint* data across seven resynthesis families against competing predictors, with permutation controls (§4). (v) A calibrated operational evaluation of deployed watermarks at matched perceptual quality, separating *operational failure* (sign/order-inverted scores under the original rule, thresholds from an independent calibration set) from loss of *separability* (oriented AUC) — the two are routinely conflated (§4).

2. THE ANALYSIS–RESYNTHESIS CHANNEL

A watermark embedder E_κ maps a host $x \sim P_0$ to $E_\kappa(x)$ (additively, $E_\kappa(x) = x + \delta_\kappa$), giving the watermarked law P_1. A detector knowing κ tests $H_0 : P_0$ vs $H_1 : P_1$; with N i.i.d. looks the optimal Bayes error decays as $e^{-N\,C(P_0,P_1)}$, C the Chernoff information [14, 15]. A *blind launderer* is

$$\mathcal{W} \;=\; \mathcal{S} \circ \mathcal{A}, \tag{1}$$

where $\mathcal{A}$ maps a waveform to a representation z (a mel envelope, a codec latent) and $\mathcal{S}$ regenerates a waveform from z with randomness independent of the input. The attacker knows neither κ nor the scheme (adaptive, key-aware removal [16] is out of scope). Imperceptibility is a budget $\rho(\delta; x) \le D$.

Proposition 1 (general channels)**.** *For any channel $\mathcal{W}$, $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) \le C(P_0, P_1)$. If moreover $\mathcal{A}(E_\kappa(x)) = \mathcal{A}(x)$ holds P_0-a.s. and $\mathcal{S}$ depends on its input only through z, then $\mathcal{W}_\# P_1 = \mathcal{W}_\# P_0$; hence $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) = 0$ and no detector beats chance at any sample size.*

Proof. The first part is the DPI for the Chernoff family: each $-\log \int p_0^{1-s} p_1^s$ is monotone under Markov kernels and the max over s preserves monotonicity [15]. For the second, $\mathcal{W}(E_\kappa(x)) = \mathcal{S}(\mathcal{A}(E_\kappa(x)), U) = \mathcal{S}(\mathcal{A}(x), U) = \mathcal{W}(x)$ a.s. with $U \perp x$, so the output laws coincide. $\square$

Proposition 1 is all we assert for real nonlinear vocoders and codecs: laundering cannot help the detector, and *exactly* analysis-invariant embeddings are *exactly* erased. It supplies no constants

Code, proofs, per-clip score arrays, and locked environments: `github.com/appleweiping/resynthesis-watermark-limits`.

and no prediction of partial survival. Quantitative structure requires a model.

3. LINEAR-GAUSSIAN SURROGATE

Work in whitened coordinates, $P_0 = \mathcal{N}(0, I_n)$, perceptual weighting carried by a masking metric $M \succ 0$ ($\rho(\delta) = \delta^\top M\delta$). Let $\mathcal{A} \in \mathbb{R}^{k\times n}$ have full row rank, $P = \mathcal{A}^+\mathcal{A}$ the orthogonal projector onto $\mathrm{row}(\mathcal{A})$, and let synthesis resample the complement from the clean prior:

$$\mathcal{W}(x) = Px + (I-P)w, \qquad w \sim \mathcal{N}(0, I_n) \text{ fresh.} \tag{2}$$

Theorem 1 (exact surviving shift and exponent)**.** *Under* (2)*, for a fixed additive shift δ: $\mathcal{W}_{\#}\mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$ and*

$$C(\mathcal{W}_{\#}P_0, \mathcal{W}_{\#}P_1) \;=\; \tfrac{1}{8}\|P\delta\|^2. \tag{3}$$

The channel transmits exactly the equivalence class $[\delta] \in \mathbb{R}^n/\ker\mathcal{A}$: the effective perturbation is $P\delta$. Three consequences, stated carefully: $\delta \in \ker\mathcal{A}$ is erased exactly (recovering Prop. 1); *mixed* perturbations survive *partially* (exponent $\frac{1}{8}\|P\delta\|^2 > 0$ whenever $P\delta \neq 0$); and row-space-only perturbations waste no budget on erased components but are *not* the unique surviving set. The budget-optimal exponent is $\max_{\delta^\top M\delta \le D} \frac{1}{8}\|P\delta\|^2 = \frac{D}{8}\lambda_{\max}(P, M)$, at the top generalized eigenvector of (P, M).

A rate lower bound (not capacity). Writing surviving shifts in an orthonormal basis of $\mathrm{row}(\mathcal{A})$, a *blind* (host-uninformed, Gaussian codebook) embedder facing a κ-aware but host-blind decoder achieves any rate below

$$R_{\mathrm{LB}} = \max_{Q\succeq 0,\ \mathrm{tr}(M_{\mathrm{row}}Q)\le D} \tfrac{1}{2}\log\det(I+Q) \tag{4}$$

nats per invariant coordinate (host as unit-variance interference; water-filling over M_{row}'s eigenmodes). R_{LB} is an *achievable lower bound for that restricted class*: informed (host-aware) embedding in the Costa/Moulin–O'Sullivan sense [12, 13] can exceed it, and we prove no rate converse. A nullspace embedding has $R_{\mathrm{LB}} = 0$ after $\mathcal{W}$.

From surrogate to real channels. For a real attacker $\mathcal{W}$ the model suggests a *measurable* channel-relative predictor: the attacker's own analysis sensitivity

$$s_{\mathcal{W}}(\delta; x) = \|\mathcal{A}_{\mathcal{W}}(x+\delta) - \mathcal{A}_{\mathcal{W}}(x)\|/\|\delta\|, \tag{5}$$

computed with *that* attacker's $\mathcal{A}_{\mathcal{W}}$ (a mel front-end, a codec encoder). Whether $s_{\mathcal{W}}$ predicts post-attack separability is an *empirical hypothesis* — the surrogate motivates it but does not prove it. Testing it out-of-sample is E2's job; a single mel fraction is never reused across attackers with different $\mathcal{A}_{\mathcal{W}}$.

4. EXPERIMENTS

Setup. LibriSpeech 16 kHz: test = [pending] speaker-stratified 4 s voice-active clips from test-clean ([pending] speakers); detector thresholds come *only* from [pending] independent calibration clips from dev-clean (speaker-disjoint from test by partition); the predictor's mapping is fit on a dev-clean fit split and evaluated on test-clean only. Attackers: STFT-GL (near-lossless control), mel-GL (spectral inversion of THE unified 80-mel analysis), Vocos (neural vocoder), Vocos-EnCodec (learned reconstruction from codec tokens), EnCodec at $6/3/1.5$ kbps, DAC, SNAC, and kNN-VC self voice-conversion (top-$k \in \{4, 8\}$) — content-, speaker- and prosody-preserving re-synthesis. All formal runs fail loudly if any attacker or baseline is missing. Scores, seeds, keys, manifests, and locked environments ship with the repository.

Constructed marks (geometry probes). W.r.t. the unified mel analysis $\mathcal{A}$ we construct, per utterance and key: a *kernel mark* (alternating projections onto quadrature $\cap$ consistent spectrograms — the first-order kernel of magnitude analysis) and a *row mark* (in-phase mel-pattern shifts). Construction quality is *measured*, not assumed: at infinitesimal amplitude the kernel mark's analysis change is $[pending]\times$ a random direction's; at the operating amplitude the (second-order) leakage is $[pending]\times$ random, and the row mark's is $[pending]\times$. Their detection uses a genie-aided matched-filter *probe* (the known per-utterance perturbation) — a diagnostic of channel geometry, explicitly not a deployable detector; deployed-detector conclusions come only from the baselines.

E2 — does channel-relative sensitivity predict survival? For $\ge$100 random directions (kernel/row mixtures, random in-phase) $\times$ randomized budgets, we measure $s_{\mathcal{W}}$ (5) per attacker and the oriented post-attack AUC of the probe in *two detector domains*: waveform correlation and a fixed mel-reading matched filter. The two answer different questions — and disagree — because survival is a property of the (channel, *detector-domain*) pair: phase-rederiving attacks decorrelate waveforms while transmitting the embedded magnitude pattern, whereas waveform codecs transmit sub-mel detail the mel probe cannot see. Mappings (isotonic; $\Phi(a\sqrt{f}+b)$) are fit on dev-clean and evaluated on test-clean only. For the mel-reading detector: Spearman $[pending]$ (95% CI $[pending]$), within-attacker permutation $p = [pending]$; competitors under the identical protocol: waveform SNR ($\rho_s = [pending]$), spectral centroid ($\rho_s = [pending]$). The theory's *one-sided* implication — $s_{\mathcal{W}} \to 0$ forces chance in *every* detector domain (Prop. 1) — is tested directly: bottom-decile-$s_{\mathcal{W}}$ directions deviate from chance by $[pending]$ (mel) / $[pending]$ (waveform) median $|\mathrm{AUC} - 0.5|$, versus $[pending]$ for the rest. [Per-attacker table inserted from results.]

E1 — deployed watermarks at matched perceptual quality. AudioSeal [1], WavMark [2], and SilentCipher [3], each tuned to the same median PESQ ([pending]) before attack — no audible-vs-transparent comparisons. Per (baseline, attacker, key): raw AUC, oriented AUC (utterance-clustered bootstrap CIs), TPR at the calibration-fixed 1%-FPR threshold (Clopper–Pearson), payload accuracy, and an attack-aware recalibration diagnostic. [Table inserted from results.] The recurring pattern: under resynthesis the deployed scores often *invert* (raw AUC < 0.5) — operational failure under the original rule — while oriented separability remains substantial; codec-hardened schemes survive the codecs they were hardened against. This is precisely the detector-dependence the subspace model predicts: survival tracks where the *detector reads*, relative to what the channel preserves.

E3 — multi-bit proof of concept (no rate claim). A B-bit payload in the unified mel domain with frame-differential BPSK and a fully *blind* decoder survives mel-GL, Vocos, and codec round-trips with BER/BLER reported with binomial CIs. This demonstrates the invariant sub-channel carries bits; it is *not* an achievable-rate measurement and we claim no bits/s at target reliability.

5. RELATION TO PRIOR WORK

Attack studies document the collapse [4, 5]; invariant/ latent-aligned embedding is established practice: Latent-Mark optimizes waveforms to shift codec latents with cross-codec transfer [6], Feature-Aligned marks match speech-feature distributions [7], VoiceMark targets VC-invariant speaker latents [8]; in images, semantic watermarks realize the same prescription [9, 10, 11]. Relative to all of these we add what a method paper does not: a stated channel model with exactly-scoped guarantees (Prop. 1, Thm. 1), an explicit account of what is *not* proven (R_{LB} vs capacity [13, 12, 17]), and a falsifiable, channel-relative predictor validated out-of-sample with permutation controls. Our evaluation protocol (independent calibration, oriented-vs-raw separation) addresses measurement pathologies that affect published robustness numbers.

6. SCOPE AND LIMITATIONS

The theory covers blind, non-adaptive laundering; key-aware removal [16] is out of scope. Exact erasure requires exact analysis invariance — our constructed kernel marks realize it only to first order, with measured second-order leakage at operating amplitude. R_{LB} concerns a restricted embedder class; real capacities may be higher. The surrogate's closed forms are not claimed for nonlinear generators — for those we claim only Prop. 1 and the *measured* predictive power of $s_{\mathcal{W}}$. Survival statements are about oriented separability or calibrated operating points, never raw sub-0.5 AUC read as erasure.

7. CONCLUSION

A restricted but precise subspace model of analysis–resynthesis explains, with correctly-scoped guarantees, when speech watermarks survive laundering, and yields a measurable channel-relative predictor that generalizes across attackers and held-out speakers. Honest accounting — of what is proved, what is measured, and what deployed detectors actually do under attack — is the paper's method as much as its message.

8. REFERENCES

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